

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 11 YEARLY EXAMINATION

MATHEMATICS EXTENSION 1

2004

Time allowed: 90 minutes

Directions to Candidates

- Attempt all questions
- Start each question on a new page
- All necessary working should be shown
- Unless otherwise specified, answers must be given in their simplest form
- Approved calculators may be used in all sections.
- Use a ruler when drawing straight lines
- Marks may be deducted for careless or poorly arranged work.
- Marks shown are approximate and may be varied.

Name: _____ Class: _____

1	2	3	4	5	6	TOTAL
$\overline{12}$	$\overline{10}$	$\overline{12}$	$\overline{10}$	$\overline{9}$	$\overline{10}$	$\overline{63}$

QUESTION 1

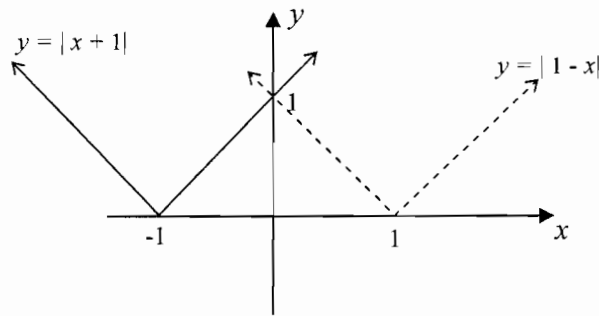
- a) If $\sin \alpha = \frac{2}{3}$ and α is acute find the exact value of $\sin 2\alpha$ 2
- b) If $x^2 - y^2 = 36$ and $x + y = 8$ find the value of $x - y$ 2
- c) Solve $2^{2x} - 6(2^x) + 8 = 0$ 3
- d) Solve $\frac{x-2}{x} \geq 1$ 3
- e) Find the exact value of $\tan \alpha$ if α is the acute angle between the lines
 $y = \frac{1}{2}x$ and $y = \frac{-1}{\sqrt{3}}x + 1$ (do not rationalize the denominator) 2

QUESTION 2

- a) Find the coordinates of the points on the curve $y = x^3 + 3x^2$ where the tangent is parallel to the line $9x - y - 5 = 0$ 3
- b) A parabola has equation $y^2 - 6y - 3 = 12x$
- i) Write the equation of the parabola in the form $(y - k)^2 = 4a(x - h)$ 2
Find the:
- ii) Coordinates of the vertex 1
- iii) Coordinates of the focus 1
- iv) Equation of the directrix 1
- c) Copy and complete the following to define a parabola as a locus 2
“A parabola is the locus of all points that.....”

QUESTION 3

- a) Show the equation of the chord PQ joining the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on the parabola $x^2 = 4ay$ is $y = \frac{1}{2}(p+q)x - apq$ 3
- b) Consider the function $f(x) = 6x^3 - 19x^2 + 11x + 6$
- i) Use the factor theorem to show that $x - 2$ is a factor of $f(x)$ 1
- ii) Completely factorise $f(x)$ 3
- c) Solve $\sin \theta - \cos 2\theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$ 3
- d) The graphs of $y = |x + 1|$ and $y = |1 - x|$ are shown 2



Hence sketch the graph of $y = |x + 1| - |1 - x|$

QUESTION 4

- a) Differentiate $y = x\sqrt{x+1}$. Give your answer as a single fraction 3
- b) Given $\cos 2A = 2 \cos^2 A - 1$ evaluate p, q and r if $\cos 4x = p \cos^4 x + q \cos^2 x + r$ 3

- c) Consider the polynomial $P(x) = (2 - x)(x + 1)^2$
- i) Sketch $y = P(x)$ indicating where it crosses the axes 2
- ii) Hence solve $P(x) > 0$ 2

QUESTION 5

- a) When the polynomial $P(x)$ is divided by $(x + 1)(x - 2)$, the result can be written as $P(x) = (x + 1)(x - 2)Q(x) + R(x)$, where $R(x) = ax + b$
- i) Given that $P(-1) = 3$, find the value of $R(-1)$ 1
- ii) Given also that the remainder is -2 when $P(x)$ is divided by $x - 2$, find the values of a and b 3
- b) i) Find the value of k if the equation $x^3 + kx^2 - 4x - 8 = 0$ has two roots equal in size but opposite in sign. 3
- ii) Hence solve the equation 2

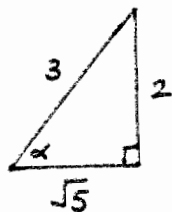
QUESTION 6

- $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$
- (a) Show that the equation of the normal to the parabola at the point P is $x + py = 2ap + ap^3$ 2
- (b) If the normal at P cuts the y -axis at Q show that the coordinates of Q are $(0, 2a + ap^2)$. 1
- (c) Show that the coordinates of R which divided the interval PQ externally in the ratio 2:1 are $(-2ap, 4a + ap^2)$. 2
- (d) Find the Cartesian equation of the locus of R . 2
- (e) Show that if the normal at P passes through a given point (h, k) then p must be a root of the equation $ap^3 + (2a - k)p - h = 0$ 2
- (f) Hence state the maximum number of normals to the parabola $x^2 = 4ay$ which can pass through any given point. 1

2004 YEAR 11 YEARLY
EXTENSION 1

Question 1

a) $\sin \alpha = \frac{2}{3}$
 $\cos \alpha = \frac{\sqrt{5}}{3}$



$$\begin{aligned}\sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ &= 2 \times \frac{2}{3} \times \frac{\sqrt{5}}{3} \\ &= \frac{4\sqrt{5}}{9}\end{aligned}$$

b) $(x+y)(x-y) = x^2 - y^2$
 $8(x-y) = 36$
 $x-y = \underline{\underline{4.5}}$

c) $2^{2x} - 6(2^x) + 8 = 0$
let $a = 2^x$
 $a^2 - 6a + 8 = 0$
 $(a-4)(a-2) = 0$
 $a = 4$ or $a = 2$
 $2^x = 4$ $2^x = 2$
 $\therefore x = \underline{\underline{2, 1}}$

d) $x^2 \times \frac{x-2}{x} \geq 1 \times x^2$

$$\begin{aligned}x(x-2) &\geq x^2 \\ x^2 - 2x &\geq x^2 \\ -2x &\geq 0 \\ x &\leq 0\end{aligned}$$

$\therefore \underline{\underline{x < 0}} \quad (x \neq 0)$

e) $m_1 = \frac{1}{2}$ and $m_2 = -\frac{1}{\sqrt{3}}$
 $\tan \alpha = \left| \frac{\frac{1}{2} - (-\frac{1}{\sqrt{3}})}{1 + \frac{1}{2} \times -\frac{1}{\sqrt{3}}} \right|$
 $= \left| \frac{\frac{1}{2} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{2\sqrt{3}}} \right|$
 $= \frac{\frac{\sqrt{3}+2}{2\sqrt{3}}}{\frac{2\sqrt{3}-1}{2\sqrt{3}}}$
 $= \frac{\sqrt{3}+2}{2\sqrt{3}-1}$

Question 2

a) $y = x^3 + 3x^2$ $m_{\text{line}} = 9$
 $\frac{dy}{dx} = 3x^2 + 6x$

$$\begin{aligned}3x^2 + 6x &= 9 \\ 3x^2 + 6x - 9 &= 0 \\ x^2 + 2x - 3 &= 0 \\ (x+3)(x-1) &= 0 \\ x &= -3, 1\end{aligned}$$

$\therefore$ points are $\underline{\underline{(-3, 0) \text{ \& } (1, 4)}}$

b) $y^2 - 6y - 3 = 12x$
i. $y^2 - 6y + 9 = 12x + 3 + 9$
 $\underline{\underline{(y-3)^2 = 12(x+1)}}$

ii. $\underline{\underline{(-1, 3)}}$ iv. $\underline{\underline{x = -4}}$

iii. $\underline{\underline{(2, 3)}}$

- c) A parabola is the locus of all points that are equidistant from a fixed point (focus) and a fixed line (directrix).

Question 3

$$\begin{aligned} \text{a) } m_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{a(p+q)(p-q)}{2a(p-q)} \\ &= \frac{1}{2}(p+q) \end{aligned}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - ap^2 &= \frac{1}{2}(p+q)(x - 2ap) \\ y - ap^2 &= \frac{1}{2}(p+q)x - ap^2 - apq \\ y &= \frac{1}{2}(p+q)x - apq \end{aligned}$$

b) $f(x) = 6x^3 - 19x^2 + 11x + 6$

i. $x-2$ is a factor if

$$f(2) = 0$$

$$\begin{aligned} f(2) &= 6(2^3) - 19(2^2) + 11 \cdot 2 + 6 \\ &= 0 \end{aligned}$$

$\therefore x-2$ is a factor

$$\begin{array}{r} 6x^2 - 7x - 3 \\ x-2 \overline{) 6x^3 - 19x^2 + 11x + 6} \\ \underline{6x^3 - 12x^2} \\ -7x^2 + 11x \\ \underline{-7x^2 + 14x} \\ -3x + 6 \\ \underline{-3x + 6} \\ 0 \end{array}$$

$$f(x) = (x-2)(6x^2 - 7x - 3)$$

$$\therefore f(x) = \underline{\underline{(x-2)(3x+1)(2x-3)}}$$

c) $\sin \theta - \cos 2\theta = 0$

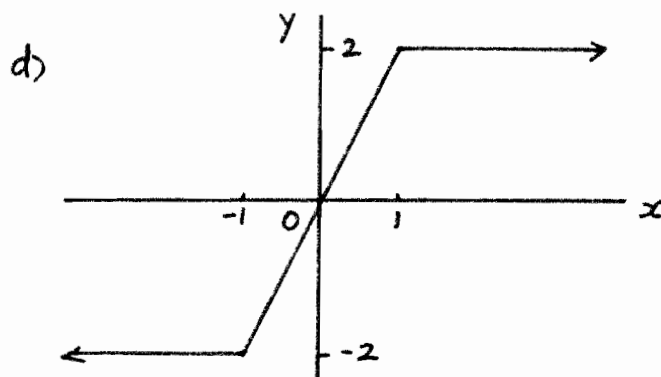
$$\sin \theta - (1 - 2\sin^2 \theta) = 0$$

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

$$(\sin \theta + 1)(2\sin \theta - 1) = 0$$

$$\sin \theta = -1 \quad \text{or} \quad \sin \theta = \frac{1}{2}$$

$$\therefore \underline{\underline{\theta = 30, 150, 270}}$$



Question 4

a) $y = x\sqrt{x+1}$

$$\begin{aligned} u &= x & v &= (x+1)^{\frac{1}{2}} \\ u' &= 1 & v' &= \frac{1}{2}(x+1)^{-\frac{1}{2}} \\ & & &= \frac{1}{2\sqrt{x+1}} \end{aligned}$$

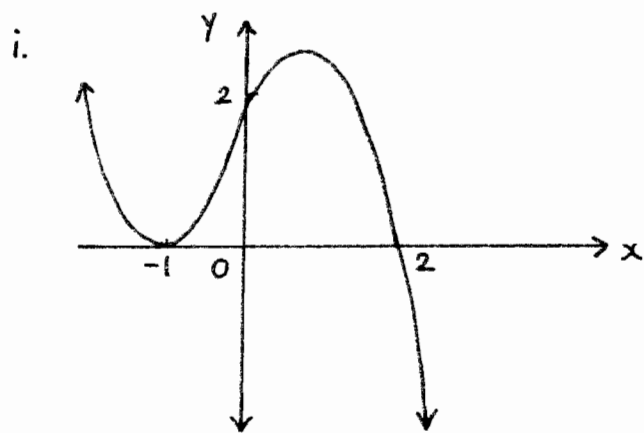
$$\begin{aligned} \frac{dy}{dx} &= \sqrt{x+1} + \frac{x}{2\sqrt{x+1}} \\ &= \frac{2(x+1) + x}{2\sqrt{x+1}} \\ &= \underline{\underline{\frac{3x+2}{2\sqrt{x+1}}}} \end{aligned}$$

$$b) \cos 2A = 2 \cos^2 A - 1$$

$$\begin{aligned} \cos 4x &= 2 \cos^2 2x - 1 \\ &= 2 (\cos 2x)^2 - 1 \\ &= 2 (2 \cos^2 x - 1)^2 - 1 \\ &= 2 (4 \cos^4 x - 4 \cos^2 x + 1) - 1 \\ &= 8 \cos^4 x - 8 \cos^2 x + 1 \end{aligned}$$

$$\therefore \underline{p=8 \quad q=-8 \quad r=1}$$

$$c) P(x) = (2-x)(x+1)^2$$



$$\begin{aligned} \text{ii. } (2-x)(x+1)^2 &> 0 \\ \underline{x < 2, \quad x \neq -1} \end{aligned}$$

$$\text{or } x < -1, \quad -1 < x < 2$$

Question 5

$$a) P(x) = (x+1)(x-2)Q(x) + R(x)$$

$$\text{i. } P(-1) = 0 + R(-1)$$

$$\therefore \underline{R(-1) = 3}$$

$$\text{ii. } P(2) = 0 + R(2)$$

$$R(2) = -2$$

$$\text{Using } R(x) = ax + b$$

$$3 = -a + b$$

$$-2 = 2a + b$$

Solve simultaneously

$$-5 = 3a$$

$$a = -\frac{5}{3}$$

$$\frac{5}{3} + b = 3$$

$$b = \frac{4}{3}$$

$$\therefore \underline{a = -\frac{5}{3} \quad b = \frac{4}{3}}$$

$$b) x^3 + kx^2 - 4x - 8 = 0$$

$$\text{i. let the roots be } \alpha, -\alpha, \beta$$

$$\text{sum of roots : } \alpha + -\alpha + \beta = -k$$

$$\beta = -k$$

$$\text{sum of roots : } -\alpha^2 + \alpha\beta - \alpha\beta = -4$$

2 at a time

$$\alpha^2 = 4$$

$$\alpha = \pm 2$$

$$\text{product of roots : } -\alpha^2 \beta = 8$$

$$-4\beta = 8$$

$$\beta = -2$$

$$\therefore \underline{k = 2}$$

$$\text{ii. } \underline{x = 2, -2}$$

Question 6

i. $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$m_{\text{tangent at } p} = \frac{2ap}{2a}$$

$$= p$$

$$m_{\text{normal at } p} = -\frac{1}{p}$$

Equation of normal :

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\underline{x + py = 2ap + ap^3}$$

ii. At Q, $x = 0$

$$0 + py = 2ap + ap^3$$

$$y = 2a + ap^2$$

$$\therefore \underline{Q(0, 2a + ap^2)}$$

iii. ratio of $-2 : 1$

$$R \left(\frac{1 \times 2ap + -2 \times 0}{-2 + 1}, \frac{1 \times ap^2 + -2(2a + ap^2)}{-2 + 1} \right)$$

$$= R \left(\frac{2ap}{-1}, \frac{ap^2 - 4a - 2ap^2}{-1} \right)$$

$$= \underline{\underline{R(-2ap, 4a + ap^2)}}$$

iv. $x = -2ap \Rightarrow p = \frac{-x}{2a}$

$$y = 4a + ap^2$$

$$y = 4a + a \left(\frac{-x}{2a} \right)^2$$

$$y = 4a + \frac{x^2}{4a}$$

$$\underline{\underline{4ay = 16a^2 + x^2}}$$

v. (h, k) satisfies equation of normal

$$h + pk = 2ap + ap^3$$

$$ap^3 + 2ap - pk - h = 0$$

$$\underline{\underline{ap^3 + (2a - k)p - h = 0}}$$

vi. 3

(a cubic can have at most three distinct solutions).